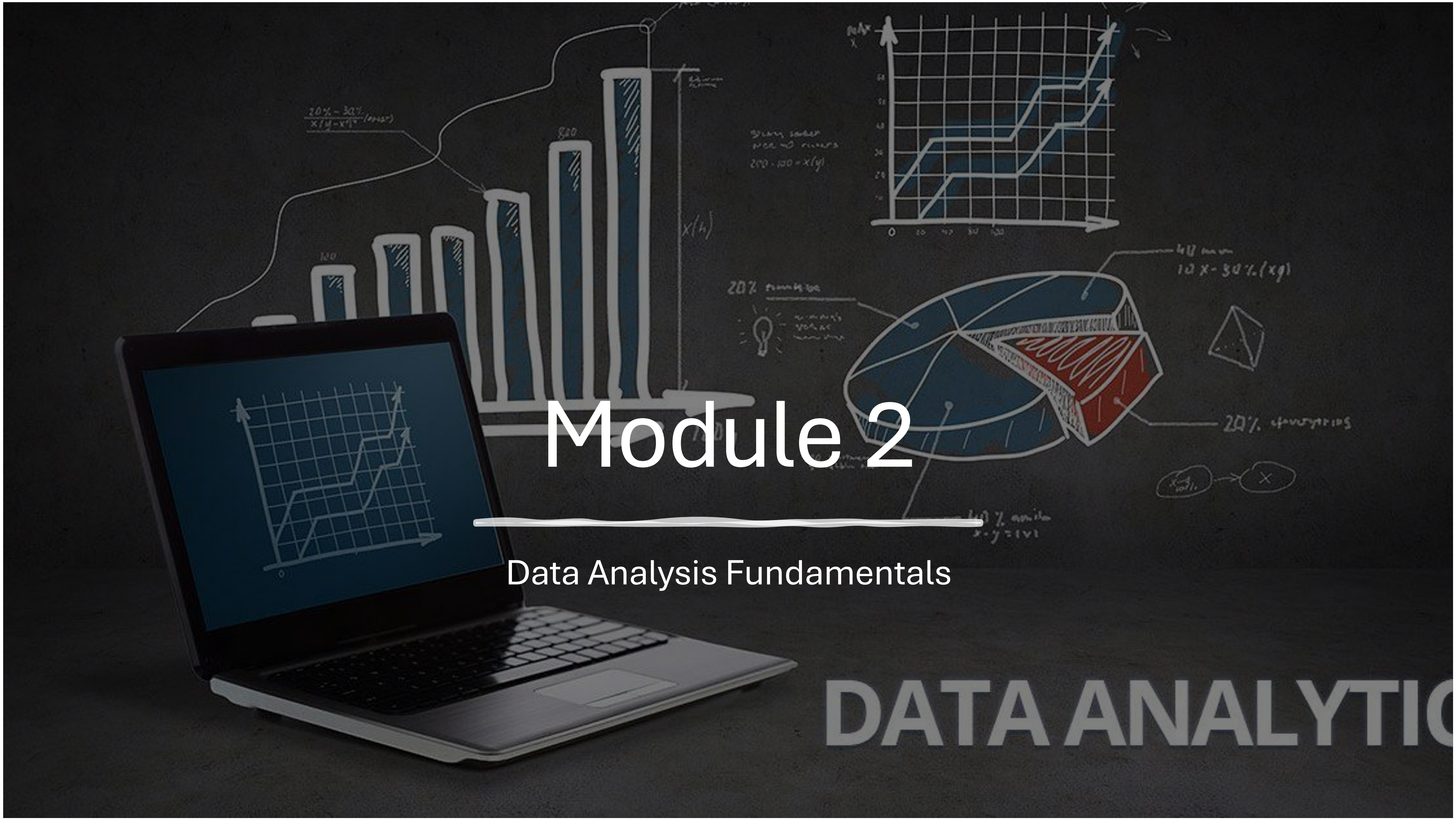
K-Youth Training Program

Data Analytics for business growth and sustainability in the workforce



Module 2

Data Analysis Fundamentals

slido

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Module Contents

Module 2: Data Analysis Fundamentals

- 1. Databases – types and purpose
- 2. Tools used to ingest data
- 3. Storing and retrieving data
- 4. Data Quality and Cleaning Techniques
- 5. Data Transformation
- 6. Data Integration and ETL Processes
- 7. Python Basics for Non-IT Students
- a. Introduction to Python and Its Uses in Data Analytics
- b. Setting Up Python Environment
- c. Basic Python Syntax and Operations
- d. Data Structures: Lists, Tuples, Dictionaries
- e. Importing and Using Libraries (NumPy, Pandas)
- f. Basic Data Manipulation with Pandas

Common Data Sources

Text

- **CSV** – Comma Separated Values
- **TSV** – Tab Separated Values
- **TXT** – Plain text format

Spreadsheets

- **Excel** – xls, xlsx
- **ODS** - Open document spreadsheet

Databases

- **SQL Database** – Oracle, MS SQL, MySQL, MariaDB...
- **NoSQL** – Mongo DB ...
- **Graph** – Neo4j, OrientDB, TigerGraph...

Markup/Metadata

- **JSON** - JavaScript Object notation - API and Data Exchange
- **XML** - Extensible Markup Language
- **YAML** - alternative for JSON - config files

Specialized

- **RDS/RData** - R data storage.
- **SAS (SAS7BDAT)** - SAS software.
- **SPSS (SAV)** - SPSS data
- **Stata (DTA)**: Used in Stata statistical software.

Big Data and Cloud

- **Parquet** – Columnar storage for big data
- **Avro** – Row based data
- **HDFS** – Big data storage

Time Series

- LOG – Elastice Search , logging data
- CDR – call data records

Image/Geospatial

- **GeoJSON –** geospatial data format
- **Shapefiles (SHP)** – GIS data
- **TIFF,PNG,JPG** – Image formats
- **PDF** – Portable Document Format data - documents

Web/API

- **HTML –** web format
- **JSON** – data exchange
- **Websocket (WS)–** direct data connections

Streaming /IoT

- **Avro/Kafka –** streaming realtime analytics
- **MQTT** – topic based events message queue

CSV

```
"product","country","date","quantity","amount","card","Cust_ID"
"prod_4","unknown","2008-12-12",1,3,,"Cust_8"
"prod_3","China","2009-04-10",2,160,"N","Cust_2"
"prod_3","China","2009-04-10",2,160,"Y","Cust_5"
"prod_3","China","2009-05-10",2,160,,"Cust_2"
"prod_3","USA","2009-05-20",20,1600,,"Cust_3"
"prod_3","Brazil","2009-06-08",15,1200,,"Cust_7"
"prod_1","USA","2009-07-04",2,70,"Y","Cust_3"
"prod_1","USA","2009-07-14",2,70,,"Cust_6"
"prod_3","USA","2009-08-20",20,1600,,"Cust_3"
"prod_2","Germany","2009-11-02",15,600,,"Cust_1"
"prod_2","Germany","2009-11-22",15,600,"N","Cust_1"
"prod_1","Germany","2009-12-02",1,35,"Y","Cust_1"
"prod_1","China","2009-12-12",1,35,"Y","Cust_2"
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"prod_1","Germany","2010-01-10",1,35,"N","Cust_1"
"prod_3","Germany","2010-01-13",1,80,,"Cust_4"
"prod_2","Germany","2010-01-15",25,1000,,"Cust_1"
"prod_2","USA","2010-01-20",2,80,,"Cust_6"
"prod_2","USA","2010-02-12",6,240,"Y","Cust_6"
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"prod_2","Brazil","2010-03-11",6,240,"N","Cust_7"
"prod_3","China","2010-03-12",1,80,,"Cust_5"
"prod_3","Germany","2010-03-14",2,160,,"Cust_9"
"prod_3","USA","2010-03-17",1,80,"Y","Cust_3"
"prod_2","Germany","2010-03-31",5,200,"Y","Cust_4"
"prod_2","USA","2010-04-22",10,400,"Y","Cust_3"
"prod_3","China","2010-05-12",2,160,"N","Cust_2"
"prod_1","USA","2010-05-17",5,175,"Y","Cust_6"
"prod_2","Germany","2010-06-22",6,240,,"Cust_1"
"prod_1","China","2010-06-28",10,350,"Y","Cust_5"
"prod_2","USA","2010-07-07",12,480,,"Cust_3"
"prod_1","Brazil","2010-07-17",5,175,,"Cust_7"
```

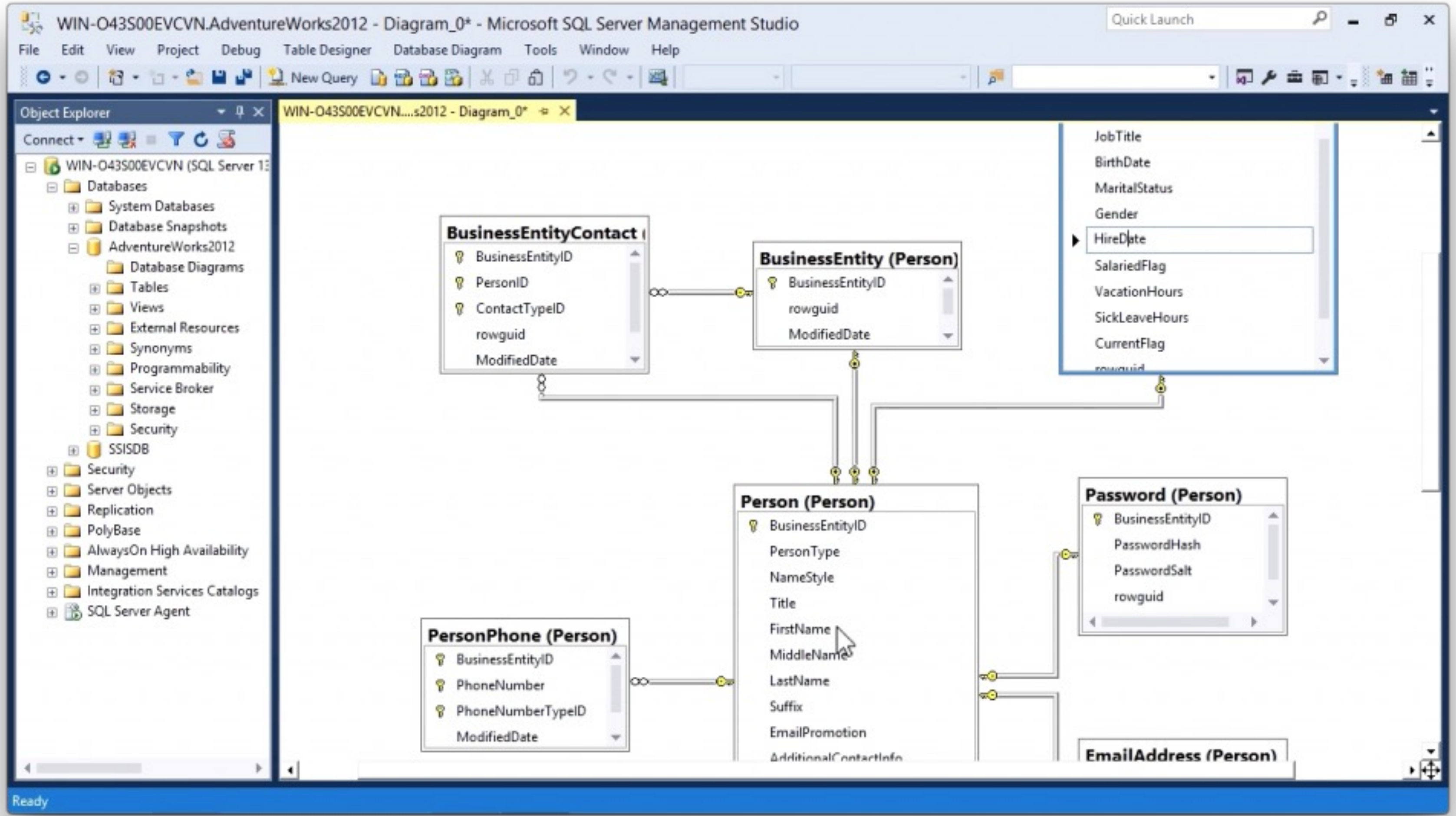
Excel

	A	B	C	D	E	F	G	H	I	J
1	Order Date	Salesperson	Dealer ID	Dealer	State	Product	Quantity	Unit Cost	Selling Price	Total Sales
2	1/1/2024	Lim Chee	C008	Cecomel Enterprise	Sarawak	Baby bathtub	40	20	33.50	1,340.00
3	1/1/2024	Lim Chee	C008	Cecomel Enterprise	Sarawak	Baby towel	50	29.90	38.50	1,925.00
4	1/1/2024	Lim Chee	C002	Lala Enterprise	Sabah	Fleece blanket	20	30	45	900.00
5	1/1/2024	Morgan	C007	Cacaca Sdn Bhd	Johor	Fleece blanket	30	30	45	1,350.00
6	1/1/2024	Morgan	C001	Merak Kayangan Enterprise	Kedah	Mittens (5 pcs/set)	20	8.90	14.90	298.00
7	1/1/2024	Lim Chee	C001	Merak Kayangan Enterprise	Kedah	Mittens (5 pcs/set)	80	8.90	14.90	1,192.00
8	1/1/2024	Zarul	C003	KukuBesi Enterprise	Pulau Pinang	One-piece pajamas	30	15.90	22.90	687.00
9	1/1/2024	Fatonah	C004	Mee Ca My Enterprise	Melaka	One-piece pajamas	100	15.90	22.90	2,290.00
10	1/1/2024	Lim Chee	C005	MyHeart Enterprise	Melaka	Rocking chair	15	115.90	180	2,700.00
11	1/1/2024	Zamira	C002	Lala Enterprise	Melaka	Romper	35	14.90	22	770.00
12	1/1/2024	Fatonah	C004	Mee Ca My Enterprise	Selangor	Romper	10	14.90	22	220.00
13	1/1/2024	Fatonah	C004	Mee Ca My Enterprise	Melaka	Romper	100	14.90	22	2,200.00
14	1/1/2024	Fatonah	C007	Cacaca Sdn Bhd	Kuala Lumpur	Romper	33	14.90	22	726.00
15	1/1/2024	Zamira	C002	Lala Enterprise	Perak	Romper	25	14.90	22	550.00
16	1/1/2024	Fatonah	C005	MyHeart Enterprise	Perlis	Shirts (5 pcs/set)	28	20	32	896.00
17	1/1/2024	Zamira	C002	Lala Enterprise	Terengganu	Shirts (5 pcs/set)	50	20	32	1,600.00
18	1/1/2024	Zamira	C007	Cacaca Sdn Bhd	Perak	Shirts (5 pcs/set)	100	20	32	3,200.00
19	1/1/2024	Fatonah	C005	MyHeart Enterprise	Perak	Shirts (5 pcs/set)	15	20	32	480.00
20	1/1/2024	Morgan	C008	Cecomel Enterprise	Pahang	Wearable blanket	16	28.90	39.90	638.40
21	1/1/2024	Zamira	C007	Cacaca Sdn Bhd	Sabah	Wearable blanket	22	28.90	39.90	877.80
22	1/1/2024	Morgan	C007	Cacaca Sdn Bhd	Johor	Wearable blanket	30	28.90	39.90	1,197.00
23	1/1/2024	Morgan	C004	Mee Ca My Enterprise	Pahang	Wearable blanket	25	28.90	39.90	997.50
24	1/2/2024	Zamira	C002	Lala Enterprise	Terengganu	Baby towel	22	29.90	38.50	847.00
25	1/2/2024	Fatonah	C010	Babyku Enterprise	Kelantan	Baby Wipes	50	8.90	14.50	725.00
26	1/2/2024	Lim Chee	C004	Mee Ca My Enterprise	N.Sembilan	Crib	2	220	330	660.00

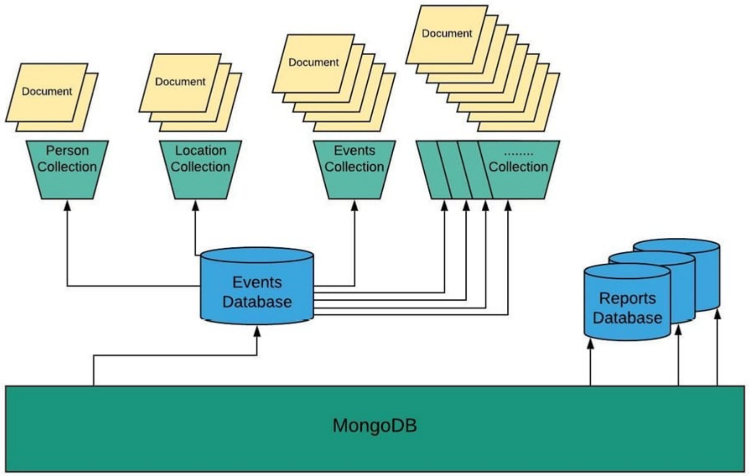
JSON


```json
[
  {
    "id": 1,
    "name": "Range Rover Sport",
    "price": "84,777",
    "url": "https://www.carhelpcanada.com/wp-content/uploads/2019/12/2020-Range-Rover-Evoque-2.jpg",
    "seats": 5,
    "miles": "14,666",
    "features": ["No Accidents", "Low KM", "Vehicle Detailed", "Leather Interior"],
    "description": "Lorem ipsum dolor sit amet consectetur adipisicing elit. Debitis odio et laboriosam!"
  },
  {
    "id": 2,
    "name": "Rolls Royce Ghost",
    "price": "455,000",
    "url": "https://robbreport.com/wp-content/uploads/2021/03/1-5.jpg?w=1000",
    "seats": 5,
    "miles": "53,666",
    "features": ["No Accidents", "Low KM", "Vehicle Detailed", "Leather Interior"],
    "description": "Lorem ipsum dolor sit amet consectetur adipisicing elit. Debitis odio et laboriosam! Numquam ut re
  },
  {
    "id": 4,
    "name": "Porsche Taycan",
    "price": "180,434",
    "url": "https://www.topgear.com/sites/default/files/cars-car/carousel/2021/02/pcgb20_1441_fine.jpg",
    "seats": 5,
    "miles": "0",
    "features": ["No Accidents", "Low KM", "Vehicle Detailed", "Leather Interior"],
    "description": "Lorem ipsum dolor sit amet consectetur adipisicing elit. Debitis odio et"
  }
]
```

Database


WIN-O43S00EVCVN.AdventureWorks2012 - Diagram_0* - Microsoft SQL Server Management Studio
Quick Launch
File Edit View Project Debug Table Designer Database Diagram Tools Window Help
New Query
Object Explorer
Connect
WIN-O43S00EVCVN.....s2012 - Diagram_0*
Databases
System Databases
Database Snapshots
AdventureWorks2012
Database Diagrams
Tables
Views
External Resources
Synonyms
Programmability
Service Broker
Storage
Security
SSISDB
Security
Server Objects
Replication
PolyBase
AlwaysOn High Availability
Management
Integration Services Catalogs
SQL Server Agent
BusinessEntityContact
BusinessEntityID
PersonID
ContactTypeID
rowguid
ModifiedDate
BusinessEntity (Person)
BusinessEntityID
rowguid
ModifiedDate
JobTitle
BirthDate
MaritalStatus
Gender
HireDate
SalariedFlag
VacationHours
SickLeaveHours
CurrentFlag
Person (Person)
BusinessEntityID
PersonType
NameStyle
Title
FirstName
MiddleName
LastName
Suffix
EmailPromotion
Password (Person)
BusinessEntityID
PasswordHash
PasswordSalt
rowguid
PersonPhone (Person)
BusinessEntityID
PhoneNumber
PhoneNumberTypeID
ModifiedDate
EmailAddress (Person)
Ready

Mongo DB – Document Database



RDBMS Relation

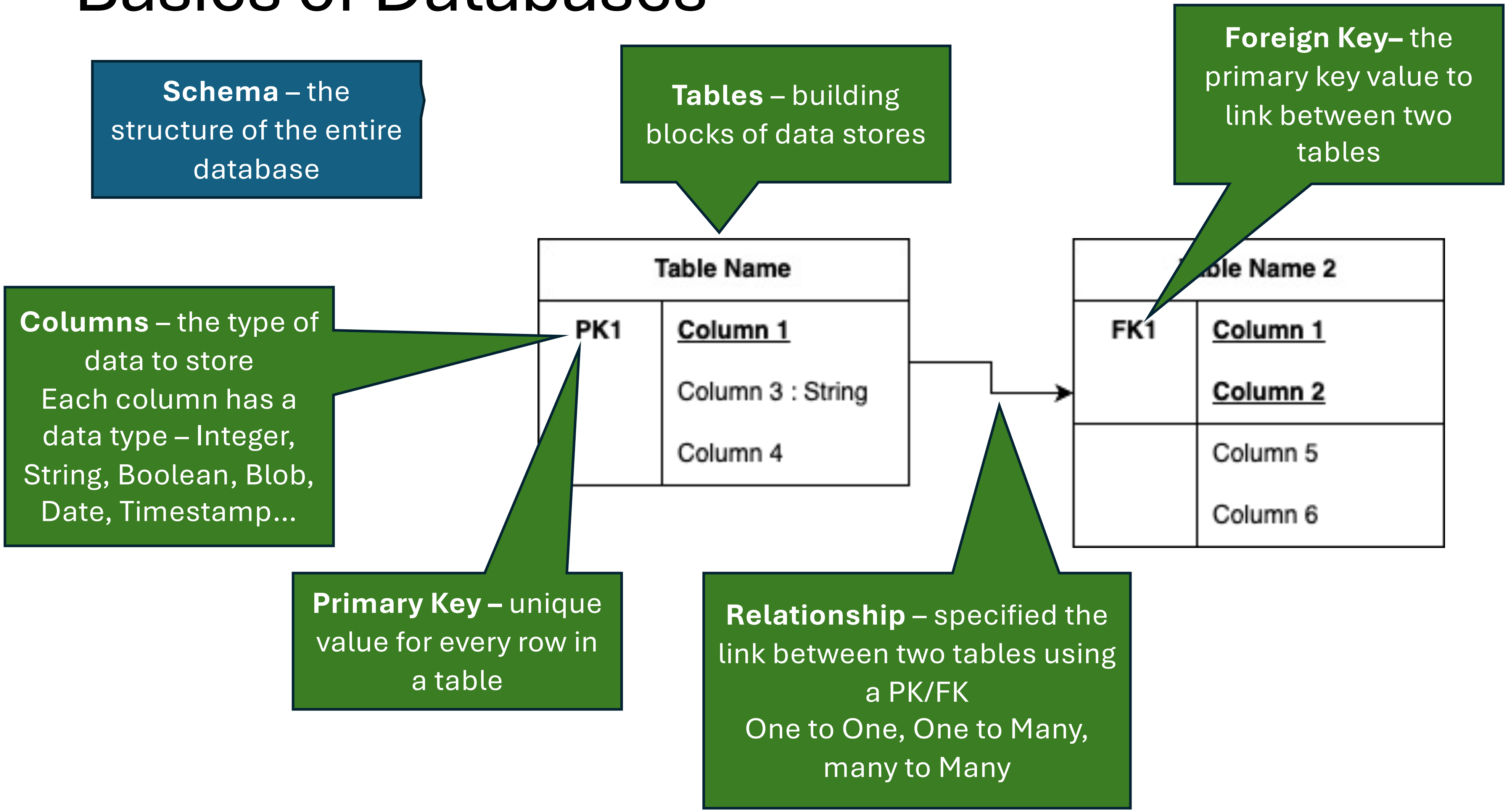
Employee Detail				
Emp_Id	**Emp_Name**	**AgencyID**	**AnnualSalary**	**GrossPay**
1	Aaron	A03031	$56705.00	$54135.44
2	Abbey	A40001	$60060.00	$59602.58
3	Abbott Cole	A90005	$70000.00	$59517.21
4	Abdal Rahim	A64120	$64365.00	$74770.82
5	Abdelmeguid	A29010	$40019.00	$16283.26
6	Abdi Ezekiel	A99070	$82780.00	$106863.56
7	Abdul Adl	A38410	$45471.00	$59418.35
8	Abdul Azi	P04002	$18408.00	$5909.64
9	Abdul Hamid	A06009	$37299.00	$35751.23
10	Abdul-Aziz,Muhammad	P04002	$20800.00	$2310.00

Employee Car Detail				
Car_Id	**Model**	**Year**	**Value**	**Emp_Id**
101	Ashok Leyland	1973	100000	1
102	Aston Martin	1965	300000	1
103	Audi	1990	100000	2
104	Bentley	1978	100000	3
105	BMW	1995	100000	4
106	Bugatti	1976	300000	5
107	Caterham	1980	200000	6
108	Chevrolet	1987	500000	7
109	Chrysler	1989	400000	8
110	Daewoo	1990	20000	9

MongoDB Documents

```
{
  "Emp_Name":"Aaron",
  "AgencyID":"A03031",
  "AnnualSalary":56705,
  "GrossPay":54135,
  "cars":[
    {
      "Model":"Ashok Leyland",
      "Year":1973,
      "Value":100000
    },
    {
      "Model":"Aston Martin",
      "Year":1965,
      "Value":300000
    }
  ]
}
```

Basics of Databases



Basic SQL

SQL (Structured Query Language) is a standard language used for accessing and manipulating relational databases.

Database Definition

1, CREATE:
```
CREATE TABLE Students (
   StudentID INT PRIMARY KEY,
   Name VARCHAR(100),
   Age INT,
);
```

Database Manipulation

1. **SELECT**: Retrieve data from one or more tables.
Example: SELECT * FROM Students;
SELECT Name from Students;

2. **INSERT**: Add new data into a table.
Example: INSERT INTO Students (StudentID, Name, Age) VALUES (1, 'Alice', 20);

3. **UPDATE**: Modify existing data in a table.
Example: UPDATE Students SET Age = 21 WHERE StudentID = 1;

4. **DELETE**: Remove data from a table.
Example: DELETE FROM Students WHERE StudentID = 1;

Convert Data to a Relational Database

Retailer	Retailer ID	voice Date	State	City	Product	rice per Uni	nits Sol	Total Sales	Operating Profit	perating Marg	Sales Method
Shoe Locker	114231	3/11/22	Selangor	Selayang	Men's Hiking Shoe	RM250.00	47	RM11,750.00	RM4,700.00	40%	Outlet
Shoe Locker	114231	3/12/22	Selangor	Selayang	Women's Running Shoe	RM210.00	67	RM14,070.00	RM4,924.50	35%	Outlet
Shoe Locker	114231	3/13/22	Selangor	Selayang	Women's Hiking Shoe	RM230.00	52	RM11,960.00	RM4,186.00	35%	Outlet
Shoe Locker	114231	3/14/22	Selangor	Selayang	Men's Casual Shoe	RM150.00	135	RM20,250.00	RM8,100.00	40%	Outlet
Shoe Locker	114231	3/15/22	Selangor	Selayang	Women's Casual Shoe	RM180.00	45	RM8,100.00	RM2,430.00	30%	Outlet
Shoe Locker	114231	3/16/22	Selangor	Selayang	Men's Running Shoe	RM220.00	62	RM13,640.00	RM6,138.00	45%	Outlet
Shoe Locker	114231	3/17/22	Selangor	Selayang	Men's Hiking Shoe	RM250.00	196	RM49,000.00	RM19,600.00	40%	Outlet
Shoe Locker	114231	3/18/22	Selangor	Selayang	Women's Running Shoe	RM210.00	161	RM33,810.00	RM11,833.50	35%	Outlet
Shoe Locker	114231	3/31/22	Selangor	Selayang	Women's Hiking Shoe	RM230.00	114	RM26,220.00	RM9,177.00	35%	Outlet
DecaShoe	114245	4/17/22	Selangor	Selayang	Men's Casual Shoe	RM150.00	77	RM11,550.00	RM4,620.00	40%	Outlet
DecaShoe	114245	4/18/22	Selangor	Selayang	Women's Casual Shoe	RM180.00	116	RM20,880.00	RM6,264.00	30%	Outlet
DecaShoe	114245	4/19/22	Selangor	Selayang	Men's Running Shoe	RM220.00	192	RM42,240.00	RM19,008.00	45%	Outlet
DecaShoe	114245	4/20/22	Selangor	Selayang	Men's Hiking Shoe	RM250.00	115	RM28,750.00	RM11,500.00	40%	Outlet
DecaShoe	114245	4/21/22	Selangor	Selayang	Women's Running Shoe	RM210.00	161	RM33,810.00	RM11,833.50	35%	Outlet
DecaShoe	114245	4/22/22	Selangor	Selayang	Women's Hiking Shoe	RM230.00	86	RM19,780.00	RM6,923.00	35%	Outlet
DecaShoe	114245	4/23/22	Selangor	Selayang	Men's Casual Shoe	RM150.00	59	RM8,850.00	RM3,540.00	40%	Outlet
DecaShoe	114245	4/24/22	Selangor	Selayang	Women's Casual Shoe	RM180.00	126	RM22,680.00	RM6,804.00	30%	Outlet
DecaShoe	114245	4/25/22	Selangor	Selayang	Men's Casual Shoe	RM150.00	152	RM22,800.00	RM9,120.00	40%	Outlet
DecaShoe	114245	4/26/22	Selangor	Selayang	Women's Casual Shoe	RM180.00	106	RM19,080.00	RM5,724.00	30%	Outlet

1. Data Repeated in Rows
2. Summation on rows

Creating and Accessing a database

Data Definition Language (DDL): CREATE, ALTER, DROP (modify the structure).

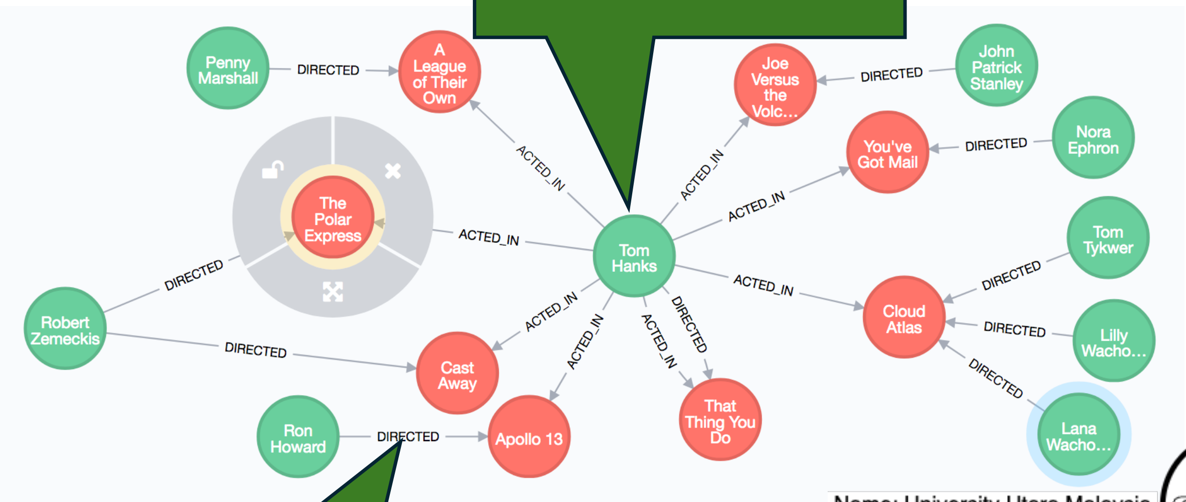
Data Manipulation Language (DML): SELECT, INSERT, UPDATE, DELETE (manipulate data).

Exercise: Convert the Excel to an Entity Relationship Diagram

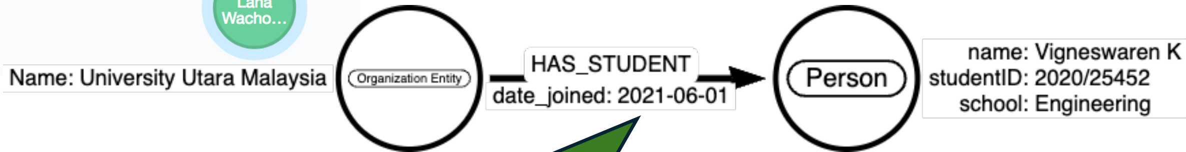
http://draw.io

Graph DB modeling

Node – represents an entity with a **Label**



Relationship – represents relationship between entities



Properties – properties of a Node / Relationship

Convert Data to a Graph Database

Retailer	Retailer ID	Invoice Date	State	City	Product	'rice per Uni	Units Sold	Total Sales	Operating Profit	'perating Margi	Sales Method
Shoe Locker	114231	3/11/22	Selangor	Selayang	Men's Hiking Shoe	RM250.00	47	RM11,750.00	RM4,700.00	40%	Outlet
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Shoe Locker	114231	3/15/22	Selangor	Selayang	Women's Casual Shoe	RM180.00	45	RM8,100.00	RM2,430.00	30%	Outlet
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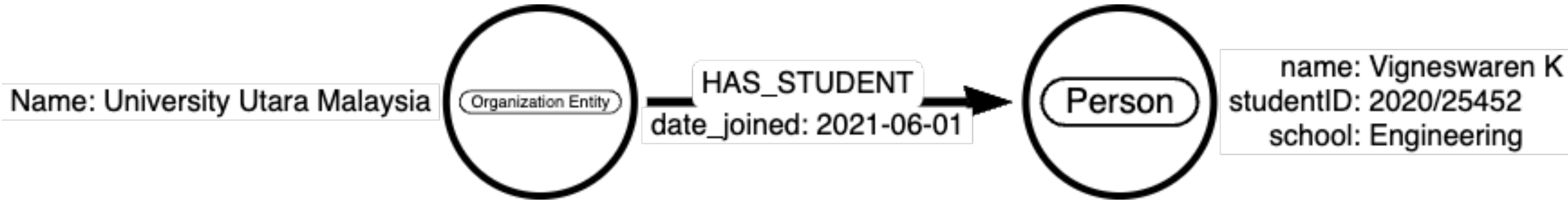
DecaShoe	114245	4/23/22	Selangor	Selayang	Men's Casual Shoe	RM150.00	59	RM8,850.00	RM3,540.00	40%	Outlet
DecaShoe	114245	4/24/22	Selangor	Selayang	Women's Casual Shoe	RM180.00	126	RM22,680.00	RM6,804.00	30%	Outlet
DecaShoe	114245	4/25/22	Selangor	Selayang	Men's Casual Shoe	RM150.00	152	RM22,800.00	RM9,120.00	40%	Outlet
DecaShoe	114245	4/26/22	Selangor	Selayang	Women's Casual Shoe	RM180.00	106	RM19,080.00	RM5,724.00	30%	Outlet



Exercise: Convert the Excel to a Graph Database

http://arrows.app

Databases Purpose – SQL Databases

Databases store data that can be retrieved and manipulated at any time.

- SQL Database (RDBMS – Relational Database Management System)
  - Data is stored in Tables, a table row contains a point of data and columns indicate the attributes of of the data.
  - Relationships of the data are indicated by Primary Keys and Foreign Keys mainly used to define the relationship between the tables and maintain the integrity of the data. Used during query to select related information
  - Uses SQL (Structured Query Language) to retrieve and manipulate data
  - Normalization is a process of reducing data redundancy and the concepts are to design the table structure and the relationships.
  - There are times when Denormalization is used to combine multiple tables in a complex schema for the purpose of reporting.
  - Applications of SQL database are found in the following use cases
    - **Retail** – for transactions and inventory, CRM
    - **Healthcare** – patient records, visits,billing
    - **Business Intelligence** — Supports data analysis and reporting in business intelligence applications.
    - **Data warehousing** – storage of vast amounts of data from multiple domains for retrieval and reporting
    - **Financial services** – transactions and customer management

Databases Purpose - Graph Database

- Graph Database
  - Data is stored as nodes(Vertex) and relationships(Edges)
  - Every node has a label and can contain properties that describe the attributes for the node
  - A relationship labels the relationship between two nodes and can also contain properties.
  - GQL (Graph Query Language) is used to retrieve and manipulate data. Cypher is the query language used for databases like Neo4j.
  - The graph excels for defining complex relationships between data and allows for large queries to execute very efficiently. (Can be 1000x faster than a RDBMS)
  - Used to explore data more effectively and visualization of patterns to gain insight of the data
  - Application of the graph include the following:
    - **Network analysis** – identify single point of failures
    - **Fraud detection –** detection of fraud patterns over large dataset
    - **Logistics –** define end to end supply chain and manage dependencies
    - **Social networking** - define complex relationship and nodes like Facebook etc
    - **Recommendation systems –** provide suggestions to users based on usage patterns like Netflix movie suggestions